

WINGS IN THE WIND

A Student's Guide to Aerofoils & The Science of Lift

Subject: Elementary Physics & Aerodynamics

Target Grade: Grades 3 - 6

1. Look Up! Introduction to Flight

Have you ever looked up into the sky and wondered how birds, airplanes, and kites can stay high in the air without falling? While they look very different, they all rely on the exact same scientific principles to defy gravity and soar through the sky! The secret lies in how air moves around specially shaped wings.

2. What is Aerodynamics?

Aerodynamics is the branch of science that studies how air moves around solid objects. Even though we cannot see air, it is made of invisible gas particles that exert force on everything they touch.

Key Concept: Forces of Air

When an object moves through the air (or when air flows past an object), the air splits to travel around it. The shape of the object determines how fast the air travels and how much pressure it creates on different sides.

3. The Shape of an Aerofoil

An **aerofoil** (or airfoil) is a specially designed wing shape that maximizes lift while minimizing drag. If you cut an airplane wing in half and look at it from the side, you will see the aerofoil cross-section:

- **Pointed / Rounded Front (Leading Edge):** Smoothly meets incoming air and splits the airflow into two streams.
- **Curved Top Surface:** Forces the air traveling over the top to take a longer path and move much faster.
- **Flatter Bottom Surface:** Allows the air traveling underneath to move in a straighter, slower path.
- **Thin Rear Edge (Trailing Edge):** Rejoins the upper and lower air streams smoothly.

4. How Does Lift Work? Step-by-Step

Lift is the upward force that holds an airplane or bird in the air. Lift is generated through a principle of physics discovered by Daniel Bernoulli: *faster-moving air creates lower pressure*.

- 1 **Faster Airflow:** Air travels **faster** over the curved top surface of the wing than underneath the flat bottom.

2 Lower Pressure Above: Because the air moves faster on top, the air pressure on top of the wing drops (becomes lower).

3 Higher Pressure Below: The slower air beneath the wing stays at a higher pressure, pushing UPWARD against the wing.

4 Lift Created! The high pressure below pushes into the low pressure area above, lifting the entire aircraft into the sky!

5. Wing Shapes: Flat Wing vs. Aerofoil Wing

Feature	Flat Wing (e.g., Simple Board)	Aerofoil Wing (e.g., Airplane Wing)
Shape Profile	Flat on top, flat on bottom	Curved top, flatter bottom, streamlined front
Air Flow Speed	Air travels at similar speeds on top and bottom	Air travels significantly faster over the top
Pressure Difference	Minimal pressure difference created	Large pressure drop above the wing
Lift Efficiency	Creates very little lift; requires steep angles	Creates high lift easily and smoothly

6. Wings All Around Us!

Aerofoils are not just designed by human engineers—nature invented them first!

- **Birds:** Birds flap their wings to create lift and thrust. Every downstroke pushes air down and back, propelling the bird upward and forward.
- **Butterflies & Insects:** Use flexible curved wings to create mini air swirls that generate lift during flapping.
- **Airplanes:** Rely on jet engines or propellers to push forward, forcing air over rigid aerofoil wings to maintain altitude.
- **Kites:** Angled against the wind to deflect air downward, generating upward force.



Hands-On Science Activity: Make Your Own Aerofoil!

Materials Needed: A small strip of paper (about 2 inches wide by 8 inches long).

Instructions:

1. Hold one short edge of the paper strip against your chin, right under your lower lip. Let the rest of the paper curve downward over your hand.

2. Blow steady, fast air straight across the TOP surface of the paper.
3. **Observe:** Instead of being blown down, the paper curve lifts UP into the air stream!
4. **Why?** Your breath created fast-moving, low-pressure air above the paper, and high pressure underneath lifted it up!

7. Summary & Key Takeaways

- **Aerodynamics** is the study of moving air around objects.
- An **aerofoil** has a curved top and flatter bottom.
- **Fast air = Low pressure.** Slow air = High pressure.
- Higher pressure underneath the wing pushes it up to create **lift**.
- Birds, insects, airplanes, and kites all use aerofoil dynamics to fly!

Student Practice Worksheet

Test your knowledge on Wings, Aerofoils, and the Science of Lift!

SECTION A 1. What is "aerodynamics"?

- A) The study of ocean currents
- B) The science of how air moves around things
- C) The study of space gravity
- D) The measure of airplane engine speed

SECTION A 2. According to Bernoulli's principle, how does air pressure change when air moves faster over a surface?

- A) Air pressure increases
- B) Air pressure decreases (becomes lower)
- C) Air pressure stays exactly the same
- D) Air pressure turns into gravity

SECTION A 3. Why does an aerofoil wing have a curved top surface?

- A) To make the airplane look modern
- B) To force air to move faster over the top
- C) To stop the air from touching the wing
- D) To make the wing heavier

SECTION B 4. Fill in the Blanks:

- a) An aerofoil wing has a _____ top and a _____ bottom.
- b) Slower air underneath the wing creates _____ pressure, which pushes the wing _____.
- c) Birds flap their wings to push air _____ and back, sending them up and forward.

SECTION C 5. Short Answer: Compare a flat wing to an aerofoil wing. Which one creates more lift and why?

SECTION C 6. Diagram Challenge: Draw a simple aerofoil wing below. Label the following parts:

- Curved Top
- Flatter Bottom
- Fast Air / Low Pressure Area
- High Pressure Area / Lift Force Arrow

